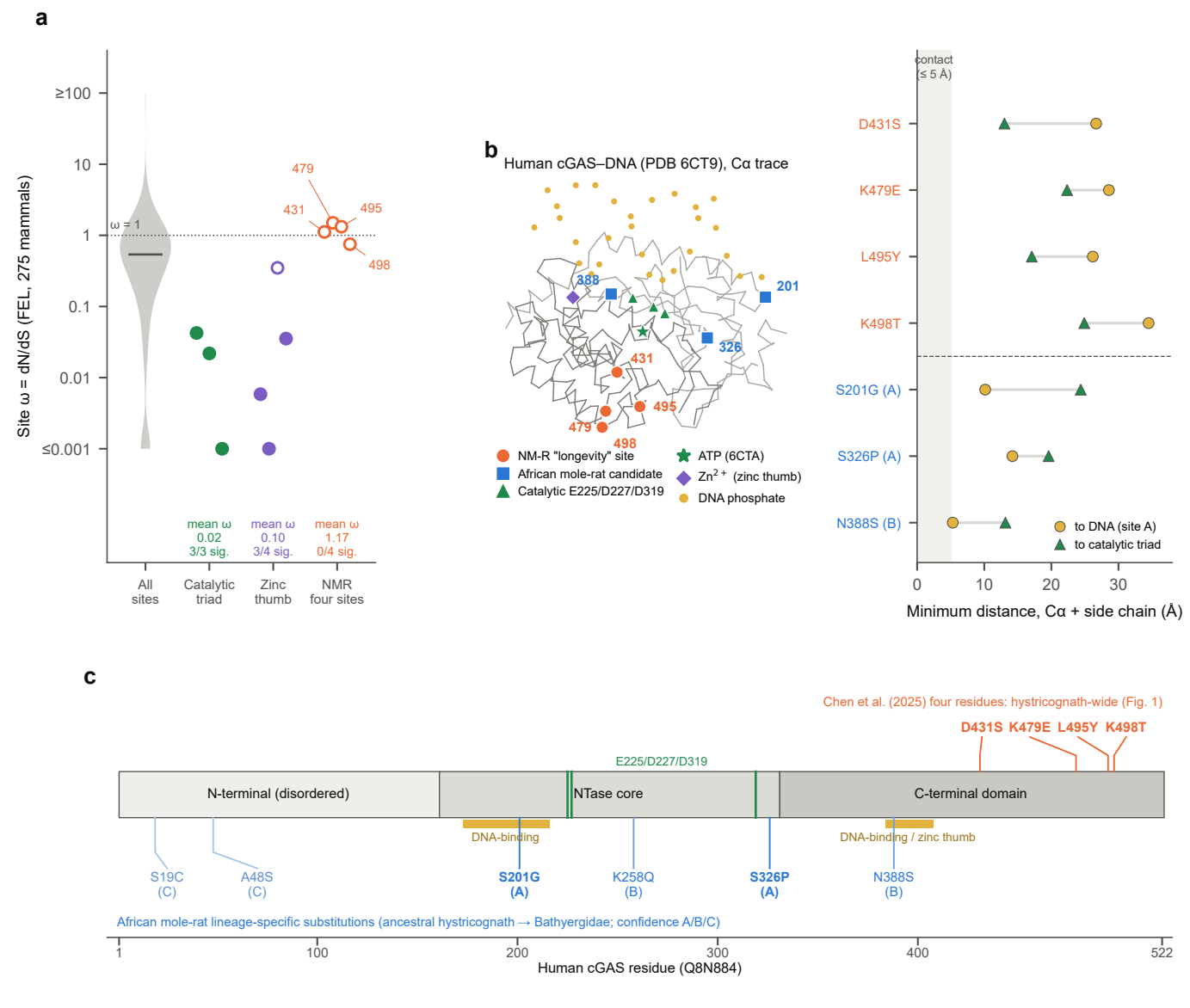

a
Site ω = dN/dS (FEL, 275 mammals)
≥100
10
1
0.1
0.01
≤0.001
ω = 1
479
431
495
498
mean ω
0.02
3/3 sig.
mean ω
0.10
3/4 sig.
mean ω
1.17
0/4 sig.
All sites
Catalytic triad
Zinc thumb
NMR four sites
b
Human cGAS–DNA (PDB 6CT9), Cα trace
388
201
326
431
495
479
498
NM-R "longevity" site
African mole-rat candidate
Catalytic E225/D227/D319
ATP (6CTA)
Zn2+ (zinc thumb)
DNA phosphate
contact (≤ 5 Å)
D431S
K479E
L495Y
K498T
S201G (A)
S326P (A)
N388S (B)
to DNA (site A)
to catalytic triad
0
10
20
30
Minimum distance, Cα + side chain (Å)
c
Chen et al. (2025) four residues: hystricognath-wide (Fig. 1)
D431S K479E L495Y K498T
E225/D227/D319
N-terminal (disordered)
NTase core
C-terminal domain
DNA-binding
DNA-binding / zinc thumb
S19C (C)
A48S (C)
S201G (A)
K258Q (B)
S326P (A)
N388S (B)
African mole-rat lineage-specific substitutions (ancestral hystricognath → Bathyergidae; confidence A/B/C)
1
100
200
300
400
522
Human cGAS residue (Q8N884)